

Kisan Vigyaan (CropShield AI): Project Synthesis

Fieldwork Documentation & Farmer Testimonials

Regions: Thanjavur, Vellore, Kallakurichi, Nagercoil

1. Thanjavur (Orathanadu Block) – Arumugam's Account

Crop Matrix: Paddy (Rice) | **Target Threat:** Brown Plant Hopper (BPH)

"Vanakkam, thambi. Let me tell you what happened during the last Samba season right here in **Orathanadu, Thanjavur**. The canals were completely full, and my paddy fields were waterlogged. Because I planted the crops closely and used plenty of nitrogen fertilizer to get a high yield, it created an incredibly thick, humid mess right down at the base of the stalks.

Suddenly, I noticed small patches of my field turning a sickly golden-brown, like the plants were being scorched by fire. It was **Hopper Burn** caused by a massive outbreak of **Brown Plant Hoppers (BPH)**. Millions of them were clustered at the bottom of the tillers, completely draining the sap.

In the past, we would just panic and blindly spray expensive chemicals all over the top of the canopy. But the spray never reached the base where the hoppers actually live! This time, the Kisan Vigyaan system flagged the extreme humidity using the field sensors and caught the stress early on satellite maps. Following the app's voice instructions in Tamil, I immediately drained the stagnant water and cleared narrow paths—what we call 'alley-ways'—through the field. That dropped the humidity and let the air circulate, which naturally broke the pest cycle and saved my rice crop from total failure."

2. Vellore (Kaniyambadi Block) – Lakshmi's Account

Crop Matrix: Groundnut (Peanut) | **Target Threat:** Tikka Leaf Spot & Wild Boars

"It has been a rough year for us in **Kaniyambadi, Vellore**. Our red soil gets incredibly dry and hard, and we rely heavily on deep tubewells. Just after a few brief rains, the blistering heat came back, and my groundnut leaves started breaking out in dark brown circular spots with bright yellow rings. It was a severe attack of **Tikka Leaf Spot**. Within days, the lower leaves were just dropping off, leaving the plants completely bare. To make matters worse, wild boars from the nearby hills were invading the fields at night, digging up the hardened soil, and eating the developing

groundnut pods right out of the earth!

Normally, we mistake these leaf spots for a lack of nutrients and throw more urea at it, which just makes the fungus grow faster. And to stop the boars, people resort to dangerous electric fencing.

Instead, the system's soil probes told us exactly when to trigger our drip irrigation so the red soil stayed soft enough for the groundnut pegs to grow safely. I uploaded a picture of the infected leaf, and the pathology engine explicitly diagnosed it as Tikka Spot, telling me the exact dose of Mancozeb fungicide to use. For the boars, we set up the smart acoustic scare system on the field boundaries. The moment it sensed movement at night, it blasted the sounds of howling hunting dogs and loud firecrackers. The boars ran right back into the hills, and we didn't have to use any dangerous wires.”

3. Kallakurichi (Chinnasalem Block) – Selvam’s Account

Crop Matrix: Hybrid Maize (Corn) | **Target Threat:** Fall Armyworm (FAW)

“Samy, farming in the dry blocks of **Chinnasalem, Kallakurichi** is always a gamble with water. We planted hybrid maize in our black cotton soil, hoping for a good harvest. But after a long dry spell, we had a sudden cloudburst. Within days, my young maize plants were absolutely devastated. The central whorls of the plants were filled with large, ragged holes and wet, messy insect droppings. The **Fall Armyworm (FAW)** caterpillars had crawled right into the heart of the plants, chewing up the growing tissue and completely destroying their ability to produce cobs.

Usually, our response is entirely reactive. We buy highly toxic chemical cocktails and manually pour them right into the heart of each plant. It’s expensive, incredibly toxic to breathe in, and by then, the inner plant is already Hollowed out anyway.

This year, the system changed our strategy. Instead of waiting for the worms to appear, the automated misting nozzles we set up were pre-programmed to spray a natural, pungent botanical mix of neem oil and Vitex negundo twice a day at dawn and dusk. This completely blocked the nocturnal moths from laying their eggs in the first place. The satellite moisture index also flagged exactly which parts of the black soil were suffering from drought stress, so we patch-irrigated those spots to give the maize plants the strength to fight off any stray larvae.”

4. Nagercoil (Agasteeswaram Block) – Mary’s Account

Crop Matrix: Coconut & Banana Plantations | **Target Threat:** Rhinoceros Beetle

“Down here in **Agasteeswaram, Nagercoil**, we get heavy rainfall from both monsoons because we are right next to the Western Ghats. It’s perfect for our coconut and Nendran banana

plantations, but the constant moisture brings unique challenges. After weeks of heavy rain, our organic manure pits became prime breeding grounds for pests. Soon after, the young fronds on my three-year-old coconut palms began to unfold, revealing horrific, completely symmetrical '**V-shaped**' cuts. A massive wave of **Rhinoceros Beetles** had flown into the crowns and bored straight into the soft apical buds to feed on the sap.

The worst part about this pest is that the damage happens hidden deep inside the high crown. By the time you see the mutilated leaves unfold at eye level, the beetle has often moved on, and the open wound rots from the rain, killing the young palm. Traditionally, we have to risk our lives climbing the wet trees to hook them out manually or pack the leaf joints with sand and naphthalene balls.

When I uploaded a photo of the damaged fronds, the Kisan Vigyaan platform immediately identified the geometric cuts as a classic Rhinoceros Beetle attack and gave me a direct audio guide in Tamil. It triggered our automated trunk-line micro-misters to release a strong-smelling camphor and neem mist directly into the crown axils during the evening hours when the beetles fly. The smell created a natural shield that kept the beetles from landing on the young buds altogether, protecting our long-term investment without anyone having to climb a single tree."